

FINANCIAL OPERATIONS ANALYTICS

Revenue Forecasting • Customer Churn • Profitability Analysis

DATA ANALYST PORTFOLIO PROJECT

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End-to-End Python Analytics Pipeline | ARIMA Forecasting | Machine Learning Churn Models | RFM & Cohort Analysis

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01 Project Overview

Financial Operations Analytics is an end-to-end data analytics project built around a subscription (SaaS-style) business. It brings together three connected analytical workstreams — revenue forecasting, customer churn prediction, and customer/segment profitability — into a single, reproducible Python pipeline that runs from raw transaction data through to a set of executive-ready outputs.

The dataset used throughout this project is synthetically generated with a fixed random seed, allowing the entire pipeline — code, data, and results — to be shared and reproduced without exposing any real company's confidential financial data. Unless otherwise noted, the metrics in this report are direct outputs of the project pipeline.

Project Snapshot

5,000 Customers Analyzed	~135K Transactions Processed	60 Months of Revenue Data	14 Visualizations Generated
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02 Business Problem & Objectives

Subscription businesses live and die by three recurring questions that finance and growth teams ask every month: how much revenue is coming next, which customers are at risk of leaving, and which customers or segments are actually worth investing in. This project builds a single analytics pipeline that answers all three from one transaction dataset.

- Forecast future monthly revenue to support financial planning and budgeting.
- Identify customers who are likely to churn before they actually cancel, and quantify the recurring revenue that is at risk.
- Segment customers using RFM (Recency, Frequency, Monetary) analysis and cohort retention to understand which customer groups are the most valuable and durable.
- Translate customer- and plan-level revenue into a profitability view that highlights where the business should focus retention and growth effort.
- Package all of the above into KPI and executive summary reports that a non-technical stakeholder can act on directly.

03 Dataset Description

The project uses a synthetically generated dataset (fixed seed, `numpy.random.seed(42)`) built to mimic a realistic five-year (2020–2024) SaaS subscription business: 5,000 customers, roughly 135,000 transactions, four subscription plans, four customer segments, eight industries and seven countries.

File	Rows	Description
financial_customers.csv	5,000	Customer master data: plan, MRR, churn status, CLV, usage & NPS
financial_transactions.csv	~135,000	Full transaction history across all customers
monthly_revenue.csv	60	Monthly revenue aggregates, 2020–2024
at_risk_customers.csv	652	Customers flagged as high churn risk by the model
rfm_segmentation.csv	4,344	RFM segments for active (non-churned) customers

Key customer-level fields include segment, industry, country, plan, MRR, contract length, number of users, support tickets, usage score, NPS score, churn status, customer lifetime, total revenue, CLV, recency, transaction count and cohort month. Full column-level definitions are documented in the project's data dictionary.

04 Data Cleaning & Preprocessing

- Parsed and standardized all date fields (signup date, churn date, transaction date, last transaction date) to a consistent datetime format for time-series and cohort work.
- Derived behavioral features per customer: recency in days since last transaction, transaction count, average transaction value, and customer lifetime in months.
- Aggregated raw transactions into a monthly revenue table (total revenue, transaction count, average transaction value, unique paying customers, and month-over-month growth).
- Engineered a cohort_month field from each customer's signup date to enable cohort retention and revenue-by-cohort analysis.
- Computed RFM quintile scores (R, F, M on a 1–5 scale) and mapped combined scores to named customer segments such as Champions, Loyal Customers, At Risk and Lost.
- Filtered churned customers out of the active-base RFM view, since RFM is intended to describe the current, still-transacting customer base.
- Verified there were no structural gaps in the 60-month revenue series before feeding it into the time-series decomposition and ARIMA modeling steps.

05 Exploratory Data Analysis

Initial exploration profiled the customer base across segment, plan, churn and usage dimensions before any modeling took place: customer distribution by segment, MRR by plan, churn rate by segment, the historical revenue trend, the CLV distribution, and usage-score patterns.

Financial Operations Analytics - Initial Data Exploration

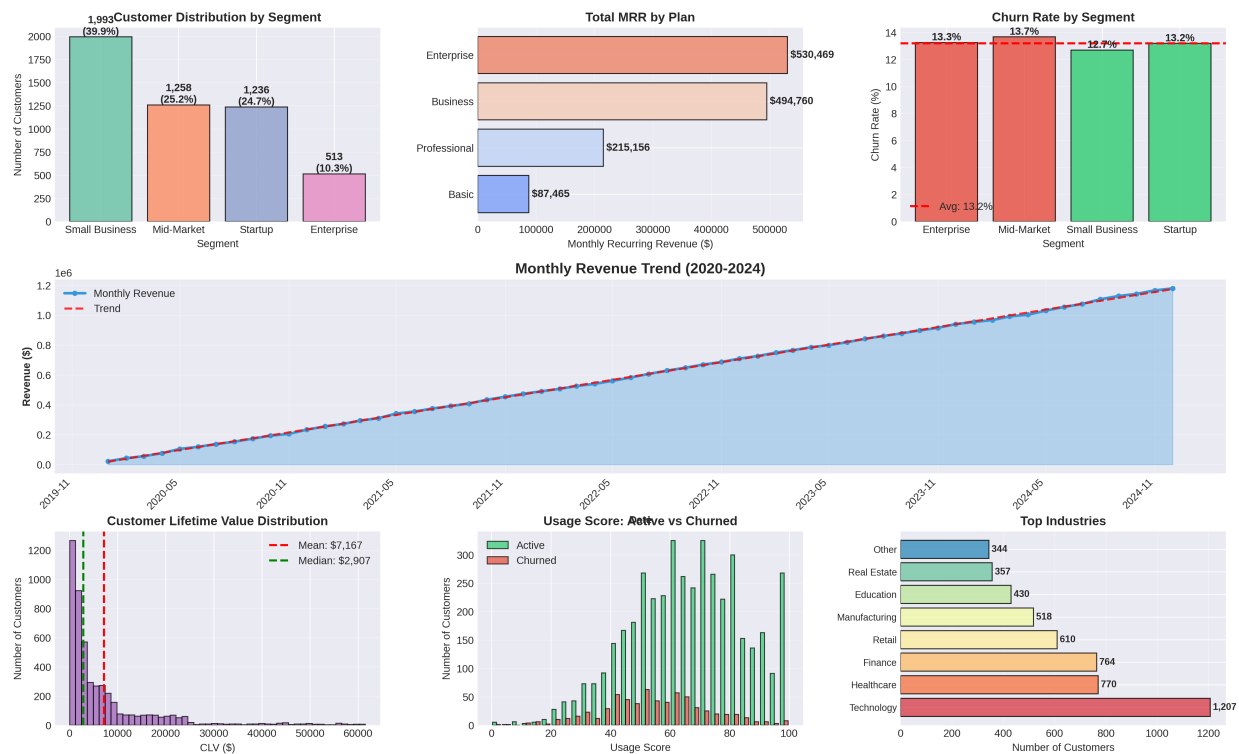


Figure 1 — Initial exploration: customer segments, MRR by plan, churn by segment, revenue trend, CLV and usage patterns.

06 Revenue Forecasting

The 60-month revenue series was decomposed into trend, seasonal and residual components to understand its underlying structure before modeling.

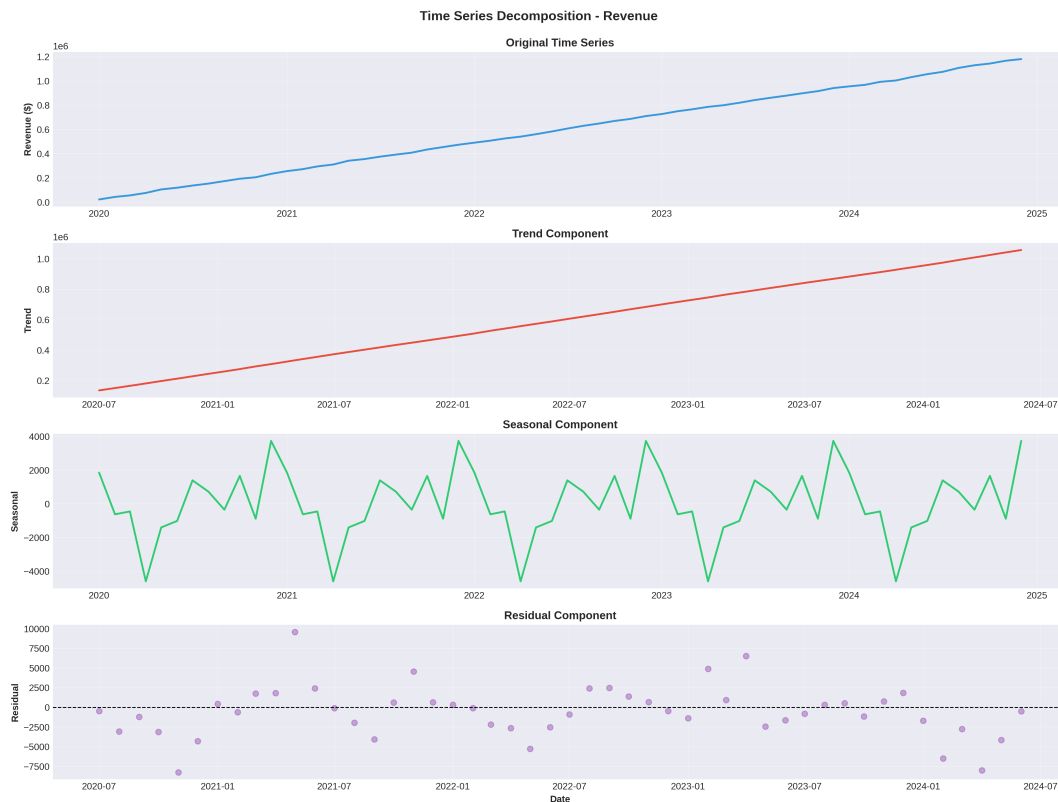


Figure 2 — Revenue trend / seasonal / residual decomposition.

Stationarity was assessed with an Augmented Dickey-Fuller test, and ACF/PACF plots on the original and differenced series guided ARIMA parameter selection.

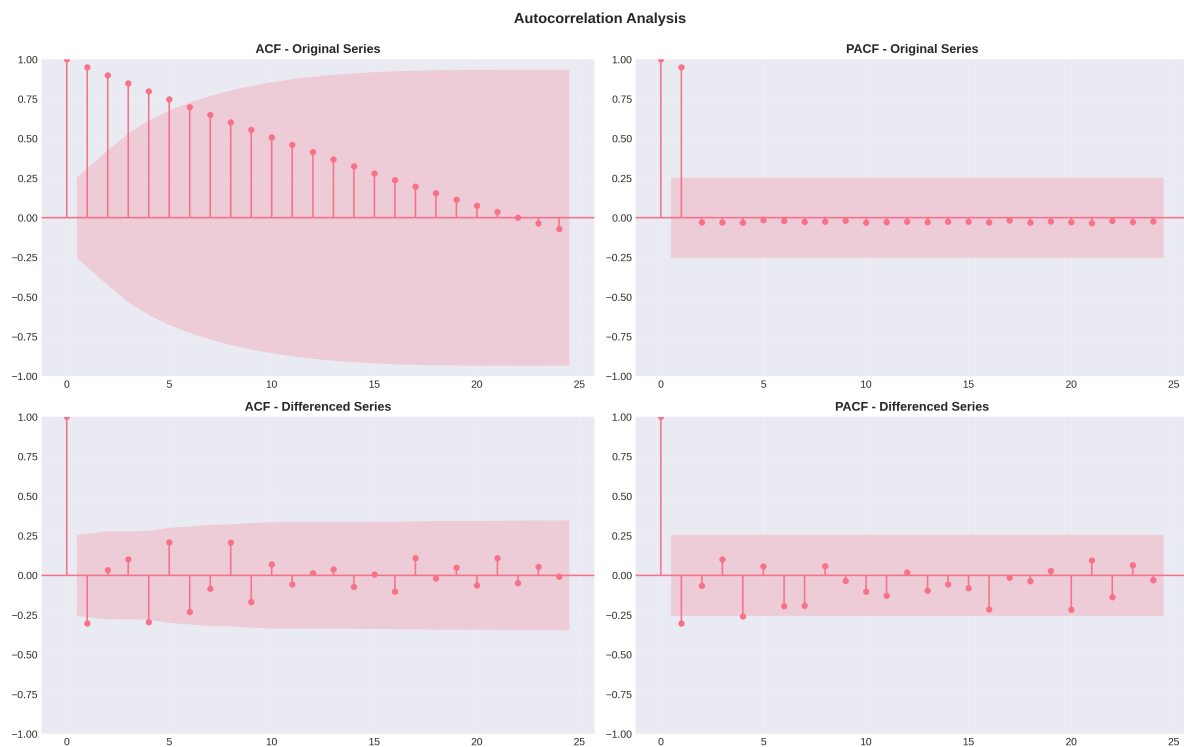


Figure 3 — ACF/PACF analysis on the original and differenced revenue series.

An ARIMA model was then fit on the historical monthly revenue and used to forecast the next 12 months with a 95% confidence interval.

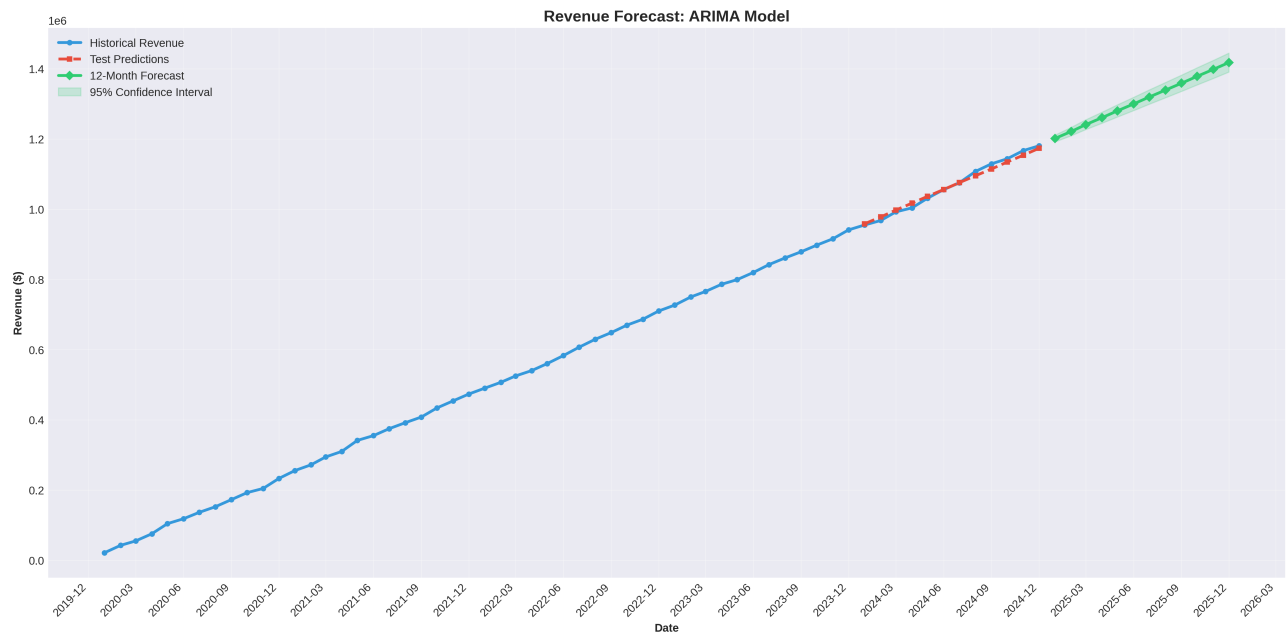


Figure 4 — ARIMA 12-month revenue forecast with confidence interval.

Forecasting Results

\$15.72M 12-Month Revenue Forecast	0.73% MAPE	\$1.31M Forecast Monthly Average	+13.26% 6-Month Revenue Growth
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A MAPE of 0.73% indicates the ARIMA model tracks the historical revenue pattern very closely over the forecast horizon, making it a reliable near-term planning input for this dataset.

07 Customer Churn Analysis

Churn behavior was first explored across segment, contract length, usage score, NPS score, support ticket volume and industry to understand where cancellations concentrate.

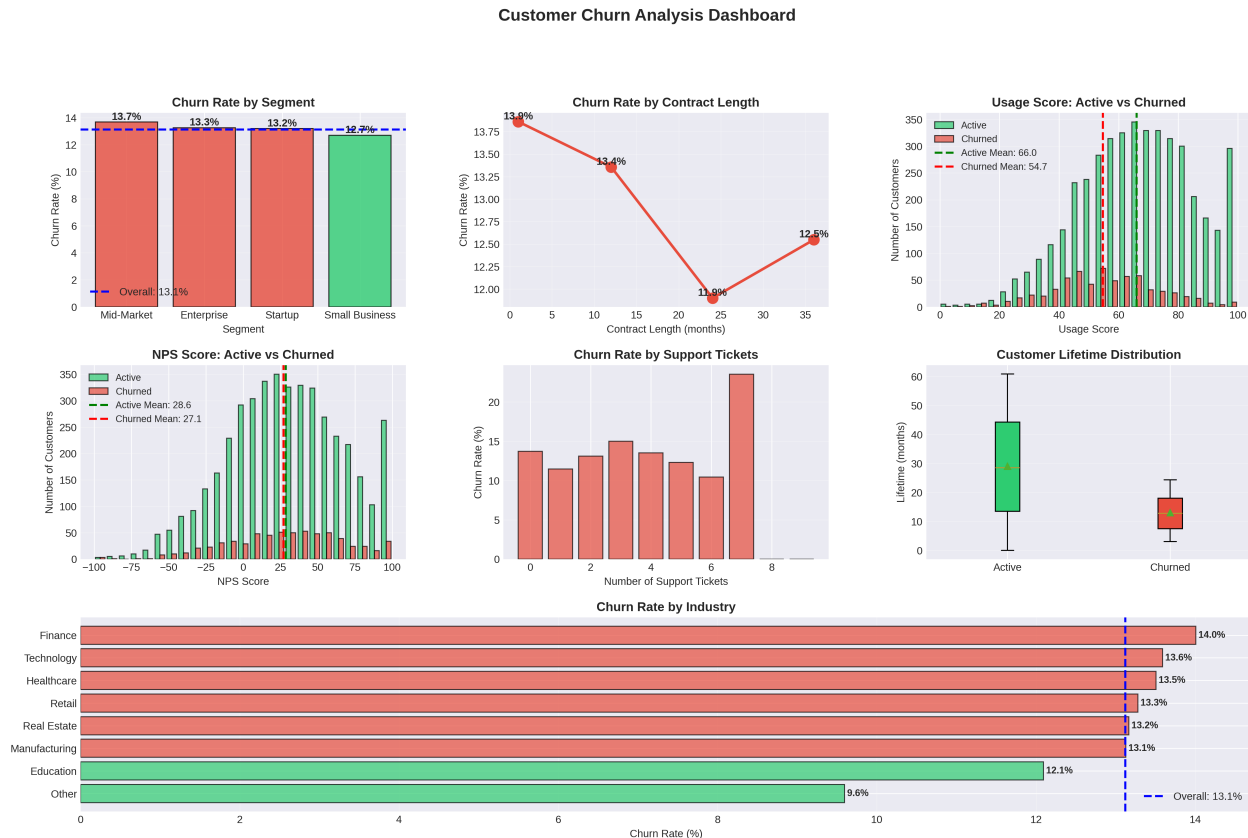
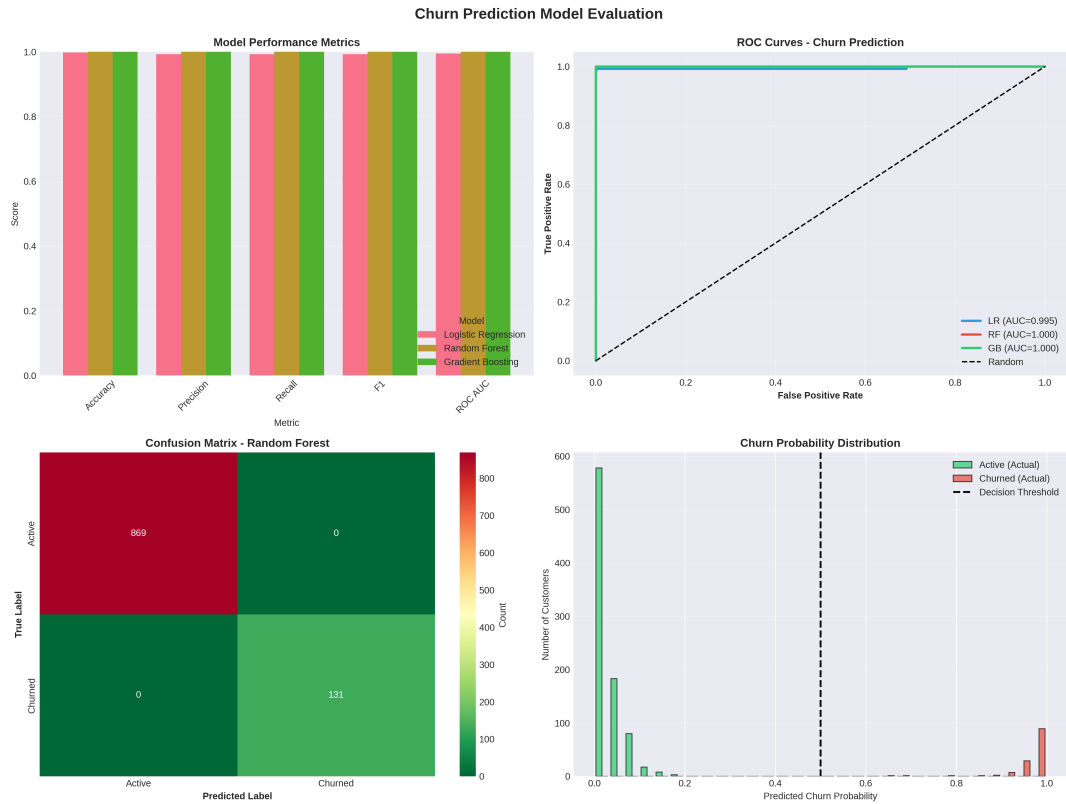
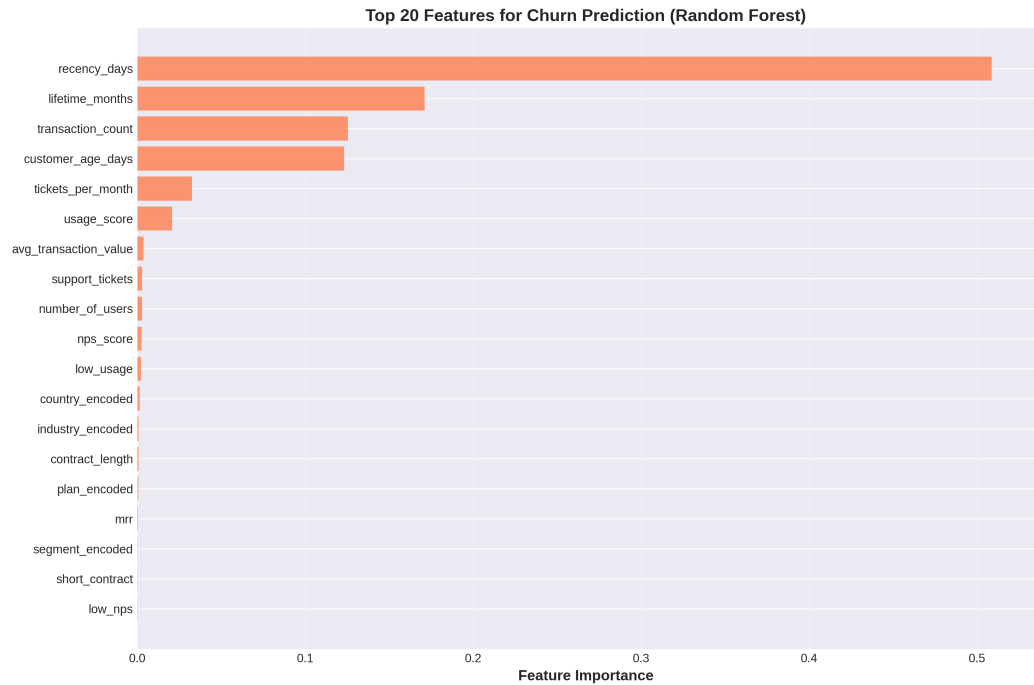


Figure 5 — Churn rate by segment, contract length, usage, NPS, support tickets and industry.

Three classifiers — Logistic Regression, Random Forest and Gradient Boosting — were trained on engineered behavioral features (recency, usage, support tickets, tenure) to predict churn, and compared using ROC/AUC and confusion matrices.



Feature importance from the Random Forest model highlighted the strongest drivers of churn in this dataset.



Predicted churn probabilities were used to stratify the customer base into risk tiers, shown below both by customer count and by MRR exposure.

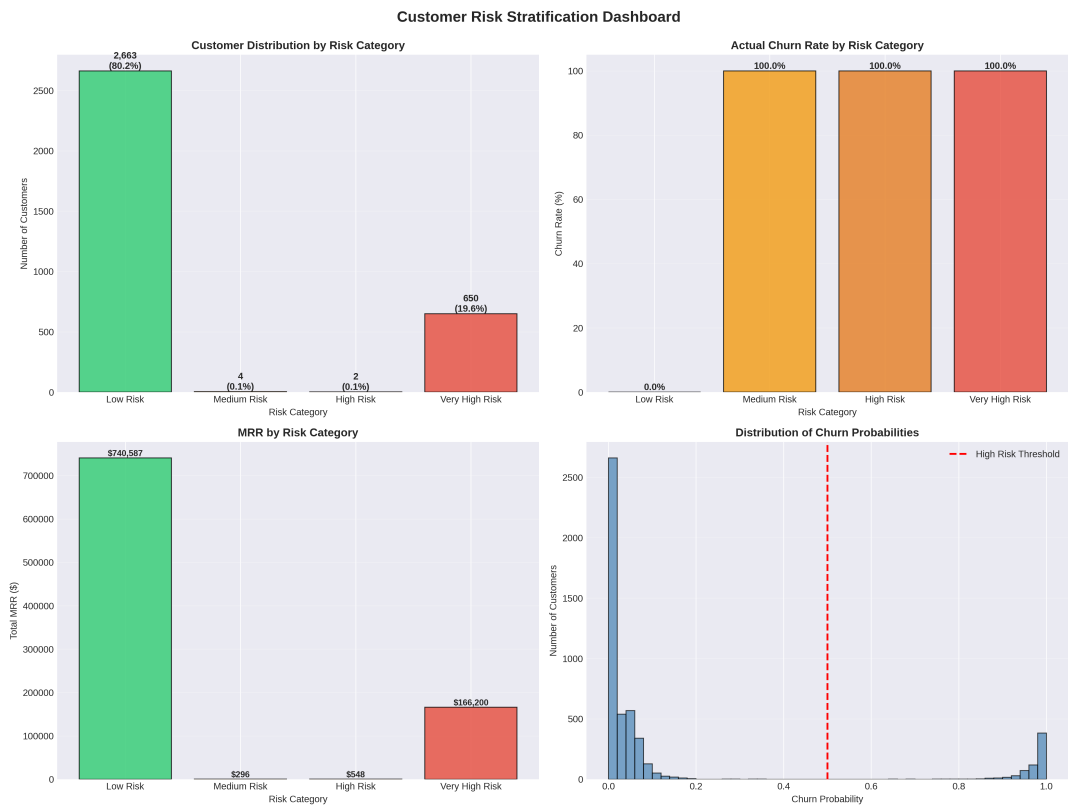


Figure 8 — Customer and MRR distribution by risk category.

Churn Results

13.12%

Overall Churn Rate

656

Churned Customers

652

At-Risk Customers (>50%)

\$166.7K

At-Risk MRR

Recency, customer lifetime and transaction count emerged as the strongest predictors of churn. The ensemble models (Random Forest / Gradient Boosting) achieved near-perfect separation between active and churned customers (ROC AUC close to 1.0) on this dataset. As documented in the project's own methodology notes, this reflects the fact that churn in the synthetically generated data is a near-deterministic function of a small number of features by design — it validates that the modeling pipeline works correctly, but should not be read as a real-world accuracy benchmark.

08 RFM & Customer Segmentation

The active customer base (4,344 customers) was scored on Recency, Frequency and Monetary value, each on a 1–5 quintile scale, and mapped into named segments to guide differentiated retention and growth strategies.

RFM Segmentation Dashboard



Figure 9 — RFM segment sizes and characteristics.

Segment	Customers	Share of Active Base
Loyal Customers	780	18.0%
At Risk	730	16.8%
New Customers	707	16.3%
Promising	632	14.6%
Champions	470	10.8%
Lost	405	9.3%
Need Attention	374	8.6%
Hibernating High Value	246	5.7%

Customer Lifetime Value (CLV) was analyzed alongside RFM to understand the distribution of long-term customer worth and its key drivers.

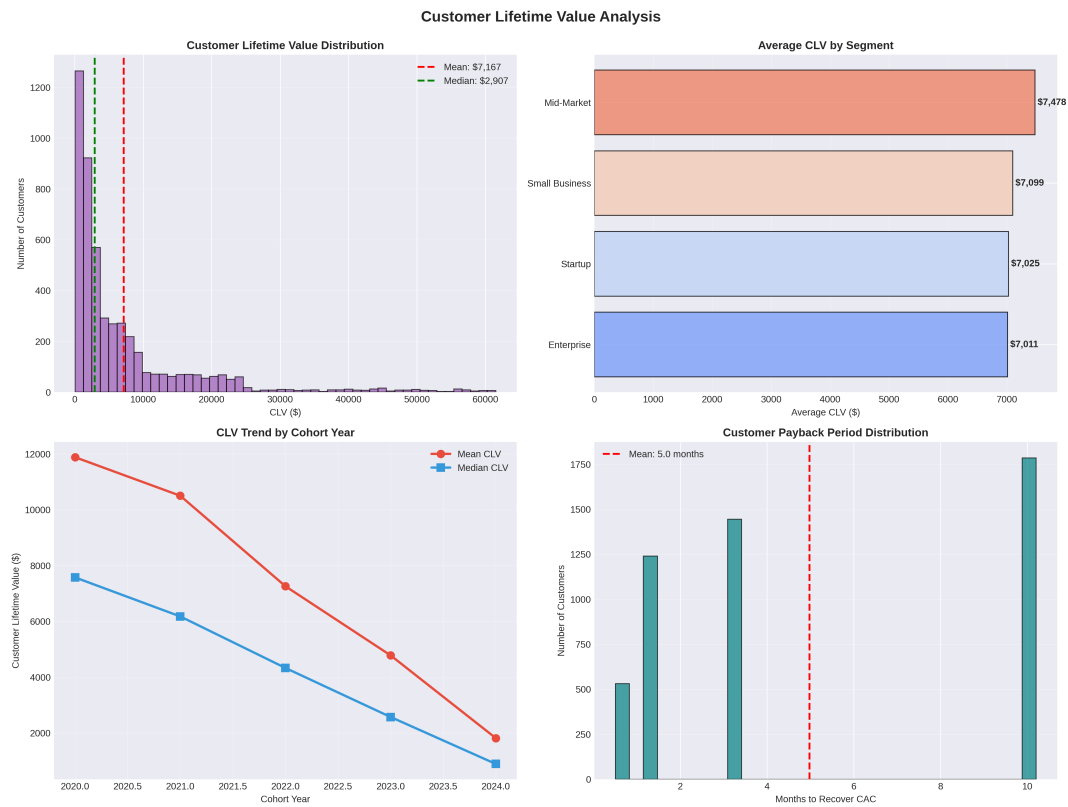


Figure 10 — CLV distribution and drivers.

Customer Value Results

\$7,166.83 Average CLV	\$2,907.19 Median CLV	14.33x CLV : CAC Ratio*	5.0 mo Avg. Payback Period
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*CLV:CAC ratio is calculated against an assumed customer acquisition cost (CAC) of \$500, used as a standard illustrative benchmark since acquisition spend is not part of the underlying dataset.

09 Cohort Analysis

Customers were grouped into monthly signup cohorts to track how retention evolves over each cohort's lifetime, and how much revenue each cohort continues to generate over time.

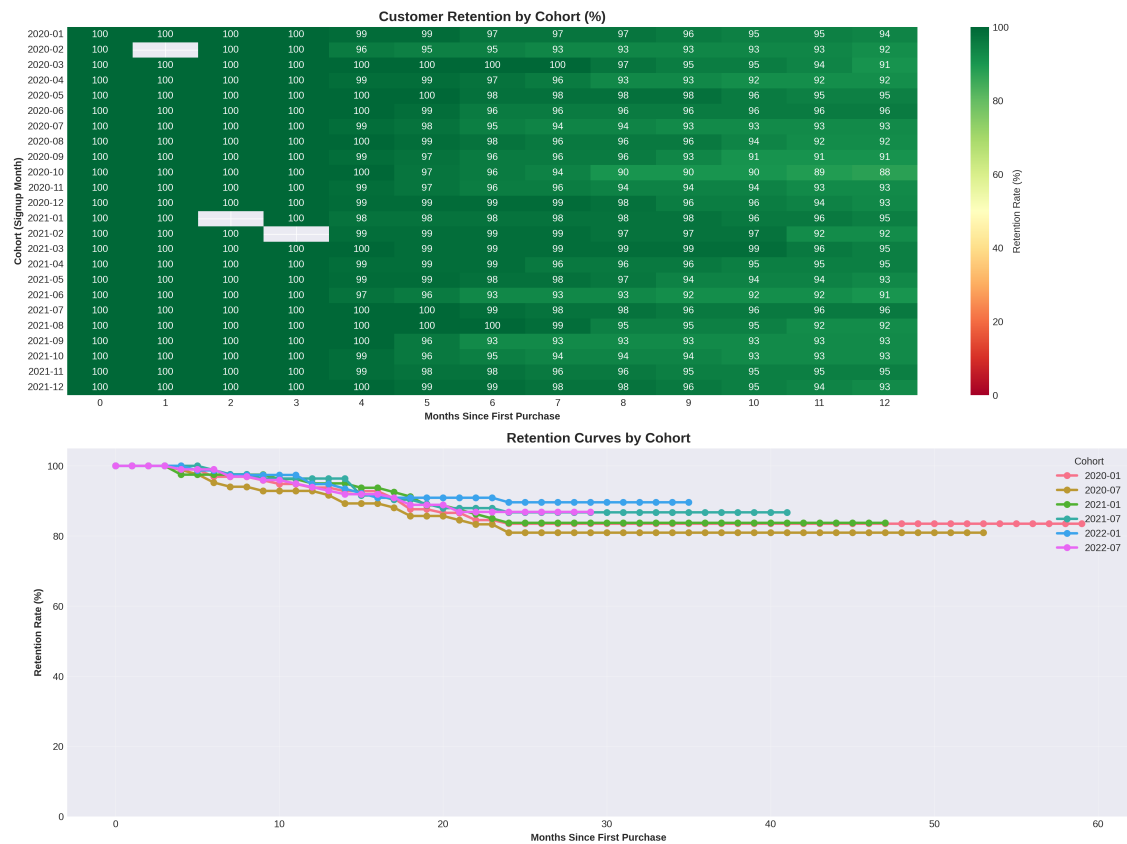


Figure 11 — Retention curves by signup cohort.

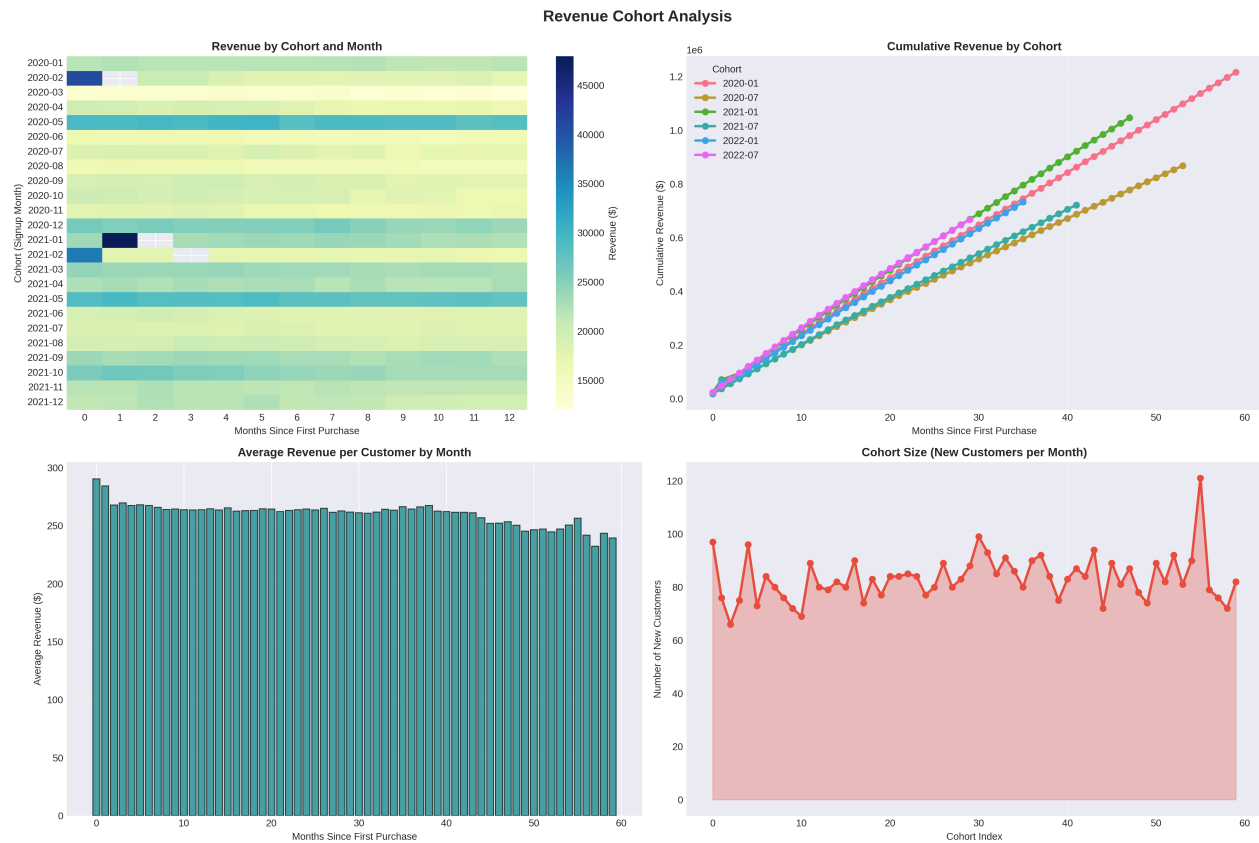


Figure 12 — Revenue generated by cohort over time.

Average customer lifetime across the base was 26.9 months, giving a practical, data-derived baseline against which the retention curves for individual cohorts can be compared.

10 Profitability Analysis

Revenue, CLV and churn were rolled up by customer segment and subscription plan to identify which parts of the business carry the strongest lifetime value and retention profile, consolidated into a single profitability dashboard.

Profitability & Segmentation Dashboard

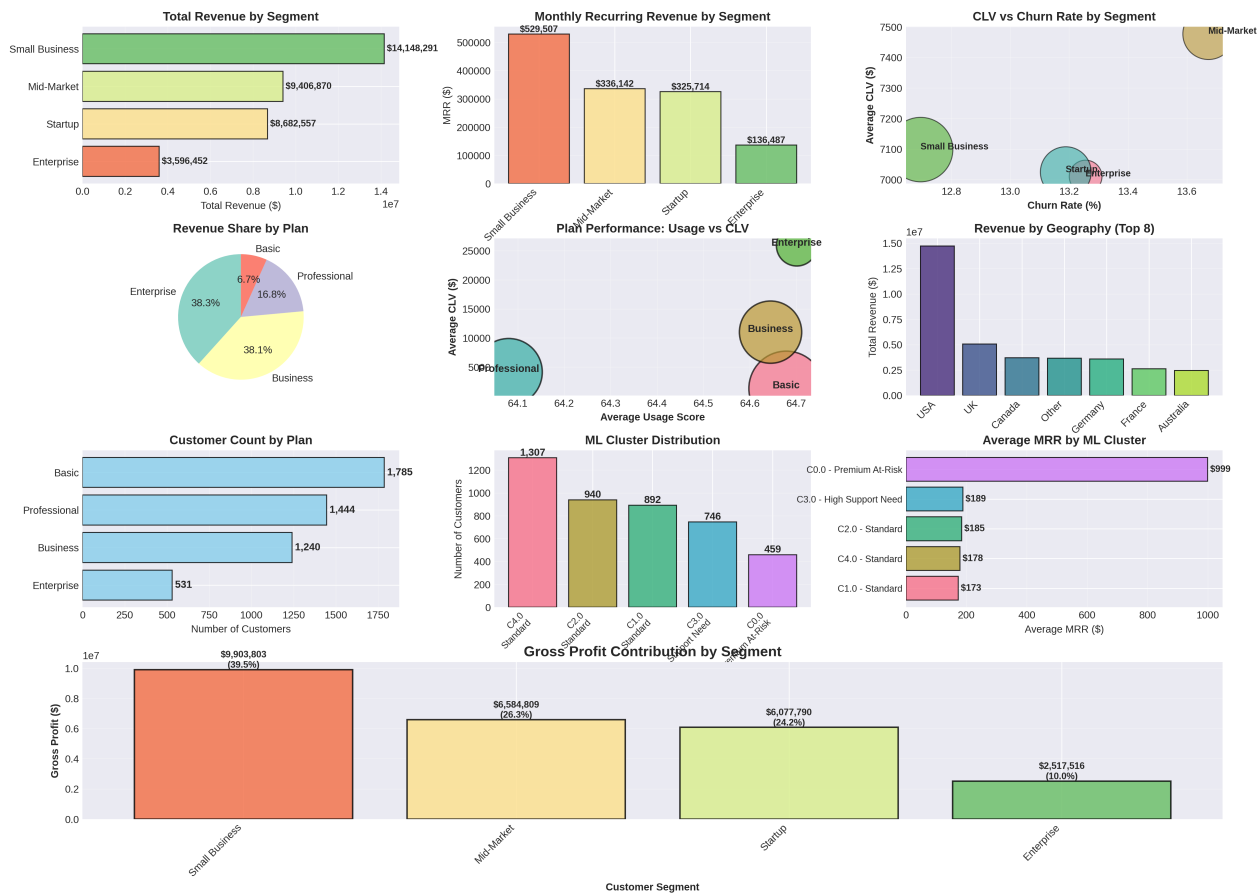


Figure 13 — Profitability by customer segment and subscription plan.

All of the above workstreams — forecasting, churn, RFM, cohorts and profitability — are brought together in a single one-page executive dashboard, the final output of the pipeline.

Financial Operations Analytics - Executive Dashboard



Figure 14 — Final executive summary dashboard.

11 Key Performance Indicators

Revenue Metrics

Metric	Value
Total All-Time Revenue	\$35,834,169.55
Current Monthly Recurring Revenue (MRR)	\$1,160,806.00
Annual Recurring Revenue (ARR)	\$13,929,672.00
Average Revenue per Account	\$7,166.83
Revenue Growth Rate (6-month)	+13.26%

Customer & Value Metrics

Metric	Value
Total Customers	5,000
Active Customers	4,344
Churned Customers	656
Overall Churn Rate	13.12%
Average Customer Lifetime	26.9 months
Average CLV	\$7,166.83
Median CLV	\$2,907.19
CLV : CAC Ratio (assumed CAC \$500)	14.33x
Average Payback Period	5.0 months

Forecasting & Risk Metrics

Metric	Value
Forecasted Revenue (Next 12 Months)	\$15,715,279.91
Expected Monthly Average	\$1,309,606.66
MAPE	0.73%
At-Risk Customers (>50% churn probability)	652
At-Risk MRR	\$166,748.00
Potential Annual Revenue Loss	\$2,000,976.00

12 Key Findings

Revenue is forecastable with high precision on this dataset.

An ARIMA model fit on five years of monthly revenue achieved a MAPE of 0.73%, indicating the 12-month, \$15.72M forecast is closely aligned with the historical pattern and can be used as a near-term financial planning baseline.

Churn is concentrated and predictable.

13.12% of the customer base (656 of 5,000 customers) has churned. Ensemble classifiers separated active from churned customers with near-perfect accuracy on this data, with recency, customer lifetime and transaction count as the leading predictors.

A meaningful slice of revenue is exposed to near-term churn risk.

652 customers were flagged with a greater than 50% predicted churn probability, representing \$166,748 in monthly recurring revenue and roughly \$2.0M in potential annual revenue at risk if no action is taken.

The active base splits into clearly actionable value segments.

RFM segmentation of the 4,344 active customers surfaces distinct groups — Loyal Customers (780), At Risk (730), New Customers (707), Promising (632), Champions (470), Lost (405), Need Attention (374) and Hibernating High Value (246) — each warranting a different retention or growth motion.

Customer value is highly skewed.

Average CLV (\$7,166.83) sits well above median CLV (\$2,907.19), indicating a relatively small group of high-value accounts contributes disproportionately to total customer value, which is consistent with the CLV distribution in Figure 10.

Acquisition economics look healthy against an assumed CAC.

Against an illustrative CAC of \$500, the average CLV implies a CLV:CAC ratio of 14.33x with an average payback period of 5.0 months.

13 Business Recommendations

Immediate (0–30 Days)

- Prioritize retention outreach to the 652 flagged at-risk accounts, starting with the highest-MRR customers to protect the largest share of the \$166,748 at-risk MRR first.
- Feed each customer's predicted churn_probability into the CRM / customer success workflow so at-risk accounts surface automatically rather than being found manually.

Short-Term (1–3 Months)

- Build segment-specific onboarding and success playbooks based on the RFM segments (e.g., re-engagement plays for At Risk and Hibernating High Value, nurture paths for New Customers and Promising).
- Monitor usage_score and support_tickets as early-warning indicators, since both showed a clear relationship with churn in the exploratory analysis.

Long-Term (6–12 Months)

- Target a measurable reduction in churn informed by the top model-identified drivers (recency, lifetime, transaction count), and track the resulting revenue impact against the ARIMA forecast baseline established in this project.
- Revisit the profitability view by segment and plan periodically to reallocate acquisition and retention investment toward the highest-CLV, lowest-churn cohorts.

14 Conclusion

This project demonstrates a complete, reproducible analytics workflow for a subscription business: cleaning and engineering features from raw transaction data, forecasting revenue with a validated ARIMA model, predicting customer churn with supervised classifiers, segmenting the customer base with RFM and cohort analysis, and consolidating the results into a profitability view and executive-ready reporting.

The pipeline ties financial planning, retention, and customer value analysis together around a single dataset, giving leadership a forecast to plan against (\$15.72M over the next 12 months), a concrete retention target (652 at-risk accounts worth \$166,748 in MRR), and a segmentation framework for deciding where to invest. It reflects the kind of end-to-end analysis a Data Analyst would be expected to deliver for a subscription-revenue business.

15 Tools & Technologies

Category	Tools / Techniques
Language & Environment	Python, Jupyter Notebook
Data Handling	Pandas, NumPy
Visualization	Matplotlib, Seaborn
Time-Series Forecasting	Statsmodels (decomposition, ADF test, ACF/PACF, ARIMA)
Machine Learning	Scikit-learn — Logistic Regression, Random Forest, Gradient Boosting, ROC/AUC evaluation
Customer Analytics	RFM segmentation, Customer Lifetime Value (CLV), cohort retention analysis, K-Means clustering
Reporting	Automated KPI summary and executive summary text generation

16 Limitations & Future Improvements

Limitations

- The underlying dataset is synthetically generated for portfolio purposes, not real company data, so absolute revenue and churn figures should be read as a demonstration of methodology rather than real-world benchmarks.
- The churn model's near-perfect ROC AUC is a property of how the synthetic data was generated (churn is a near-deterministic function of a few features by design) and is not representative of typical real-world churn-model performance.
- Prophet-based forecasting was part of the original design but is optional and was not included in this run, since it depends on an additional package that is not part of the core requirements; ARIMA forecasting, churn modeling, cohorts, RFM, CLV and profitability all run from the core requirements only.
- The CLV:CAC ratio relies on an assumed customer acquisition cost of \$500, since actual acquisition spend is not part of the transaction dataset.

Future Improvements

- Validate the pipeline against real (anonymized) subscription data to test whether the same feature set and model choices generalize outside the synthetic environment.
- Add Prophet as an alternative forecasting model and compare its accuracy against the current ARIMA baseline.
- Incorporate actual customer acquisition cost data, once available, to replace the assumed CAC used in the CLV:CAC calculation.
- Extend the churn model with additional behavioral signals (e.g., product feature usage, billing history) and monitor model drift over time as new data arrives.
- Automate the KPI and executive summary generation into a scheduled dashboard so stakeholders can track these metrics on a rolling basis rather than as a one-time report.

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